

13 Roman to Integer

Dry Run

Roman $\rightarrow$ M C M X C I V

ans $\rightarrow$ 0

1 condition

$\hookrightarrow$ if $(i+1 < n \text{ \& \& } curr < map.get(i+1))$
 $\hookrightarrow ans -= curr$

2 condition

$\hookrightarrow$

ans += curr

I

ROMAN $\rightarrow$ M C M X C I V , curr $\rightarrow$ 1000
0 1

ans $\rightarrow$ 1000 $\& \& 1000 ! < 100$

2 condition

①

M C M X C I V
↑ ↑ ↑
/ / /

res → 1000

~~900~~
~~1900~~
~~1890~~
~~1990~~
~~1889~~
1994

$100 < 1000$

① word

$1000 \leq 1000$ 2 can

$100 < 1000$ ①

$100 < 1$

$1 < 5$

→ We {i} reach to last idx

↳ i) $(i+1) < n$ & $curr < map.get(i+1)$

this
will help to ~~by~~ handle the
last case

→ $7 < 7$ → run else
false

~~ans~~

res $\rightarrow$ 1994 After the for loop.

$\rightarrow$ Code:

```
public int helper (String s) {
```

```
    Map < character, Integer> map = new  
                                HashMap<>();
```

```
    map.put ('I', 1);
```

```
    map.put ('V', 5);
```

```
    map.put ('X', 10);
```

```
    map.put ('L', 50);
```

```
    map.put ('C', 100);
```

```
    map.put ('D', 500);
```

```
    map.put ('M', 1000);
```

```
    int result = 0; int ans = 0;
```

```
    for (int i = 0; i < s.length(); i++) {
```

```
        map.get (s.charAt(i));
```

```
        int curr = map.get(i);
```

```
        if (i+1 < n && map.get(i+1) curr < map.get(s.charAt(i+1)))  
            ans += curr;
```


3 else {

ans += cur;

}

02 return ans;

}